

Introduction to Data Science

A Beginner's Guide



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OVERVIEW

The applications and adoption of data science across industries have been gaining momentum over the past decade.

A 2018 **report** of Deloitte Access Economics shows that **76%** of businesses plan to raise their spending on data analytics and data science capabilities within the next two years.

In another **study**, research and consulting firm MarketsandMarkets predicts that by the end of 2024, the data science market size will grow to \$140.9 billion from \$37.9 billion in 2019, at a CAGR (Compound Annual Growth Rate) of **30.0%**.

Key factors driving the growth of the data science market include the increasing adoption of data-centric business strategies, the growing need for actionable insights, and the dramatic emergence of innovative technologies such as Wi-Fi connectivity, sensors, and IoT (Internet of Things), which generate massive amounts of data every second.

The world today produces **2.5 quintillion bytes** of data, thanks to **306.4 billion emails**, over **5 million Tweets**, and **95 million videos and photos** that people share on Instagram every day. According to **recent estimates**, by 2020, our digital ecosystem will be approximately **44 zettabytes**, and by 2025, people will be generating about **463 exabytes** of data per day.

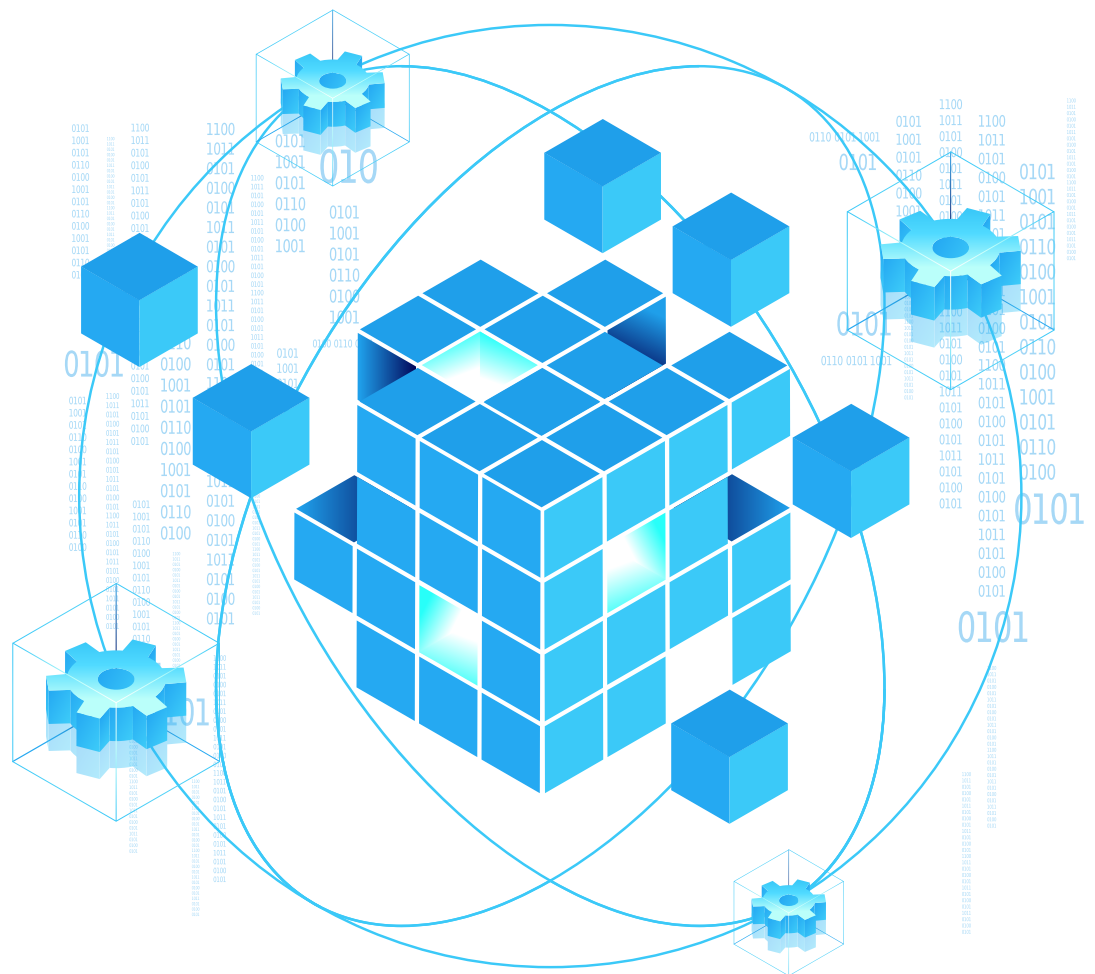
The dramatic influx and accelerated production of large-scale data, or Big Data, opens up numerous opportunities for modern organizations looking to use data to improve productivity, reduce costs, and increase profitability through business intelligence gained by processing the raw data they collect on a day-to-day basis.

A majority of companies around the world, however, are facing severe talent shortages and are struggling to fill data scientist vacancies as the supply of professionals with skill-sets needed for data science roles is far below industry demand.

Global management consulting firm McKinsey & Company **forecasts** a 50 percent supply gap in the coming years compared to the demand for data scientist professionals.

The acute talent shortfall makes data science one of the least-saturated, highly-employable, handsomely-compensated job sectors, holding abundant opportunities for aspiring data scientists seeking a rewarding career.

The **data science handbook** below will provide you with all necessary information relevant to data science, including an **introduction to data science**, its industry applications, real-life use cases, key terminologies, and the skills you need to land a job. Let's get started.



WHAT IS DATA SCIENCE?

Data science techniques involve the expertise required to collect, shape, store, manage, and analyze data for data-oriented decision making, explains Northeastern University's Professor of Data Science, Dr. Martin Schedlbauer.

The data scientist applies machine learning (ML) algorithms to audio, video, images, text, and numbers to develop AI (Artificial Intelligence) systems, which produce insights that business analysts can translate to add value to an organization's bottom line. A combination of mathematical knowledge, statistical computing, and programming skills, data science has a beneficial impact on both consumers and companies alike.

A study by the McKinsey Global Institute shows that data science can increase retailer profit margins by **60%**, and services based on location data can provide end-users with an economic surplus of **\$600 billion**.

This means that data science allows consumers to buy goods and services at a lower price than expected. For instance, if a person budgeted \$500 to buy a smartphone and then gets the same model for \$400, his/her economic surplus amounts to \$100.

Data science applications offer innumerable business benefits, and companies that are implementing the ground-breaking technology are already taking advantage of it. For example:

One hundred million dollars - that's the money Southwest Airlines Co. saved by minimizing the idle hours of its planes.

United Parcel Service, Inc. saved **9 million gallons** of fuel by optimizing its fleet, and the U.S., Internal Revenue Service, saved **2 billion dollars** by enhancing its ability to identify improper payments and fraud. portion of the prime procedures where data science has figured out how to cast its wonderful enchantment.

THE INDUSTRY APPLICATIONS OF DATA SCIENCE

Not only the technology sector, today, every industry aims to exploit data, and that is bringing data science to the forefront.

Below is a list of industries that are leveraging data science techniques and applications to drive business value.



Healthcare

By integrating machine learning algorithms, statistics, analytics, and pattern recognition, data science improves the efficiency of the healthcare industry.

A **survey** conducted by the Journal of the American Medical Informatics Association shows that the healthcare sector's demand for professionals with data science skills is surging.

Data scientists are in demand because of their inherent ability to process and analyze massive laboratory and clinical datasets using deep learning technology.

In addition, data science applications are at the core of smart healthcare wearables. This enables data scientists to quickly detect and track health issues, reducing risks, and enriching the quality of human life.



Finance

Information and numbers drive the banking and finance industry. Therefore the sector is always proactive in adopting data-driven technologies, and data science is no exception.

Data science techniques help financial institutions extract actionable insights from large data sets, promoting sustainable development and a healthy economic environment.

The role of data science in the financial sector is diverse, and it includes customer experience analysis, detection of fraudulent behavior, identifying credit or debit card misuse, personalized recommendations, and risk assessment.



Manufacturing

With the advent of Industry 4.0, the **demand for data scientists in the manufacturing sector** is hitting a record high.

As traditional industrial and manufacturing practices are getting automated at a rapid pace, the need to adopt data science is also increasing, enabling smart factory solutions.

Using AI-powered predictive analytics, data scientists are helping the manufacturing industry with preventive maintenance, such as troubleshooting potential equipment issues, which reduces delays and speeds up production.



Energy Sector

Be it exploration, production, transportation, or logistics; the energy sector often encompasses projects that are high in cost.

Because of various external factors, such as the global economic environment and political situations, the sector is also exposed to severe price fluctuations.

For this reason, leading energy companies are increasingly turning to data-driven solutions, creating significant inroads for data science applications.

By using predictive models, data scientists enable energy companies to cut costs, lower risks, reduce their downtime, and optimize investments while enhancing equipment maintenance capabilities.



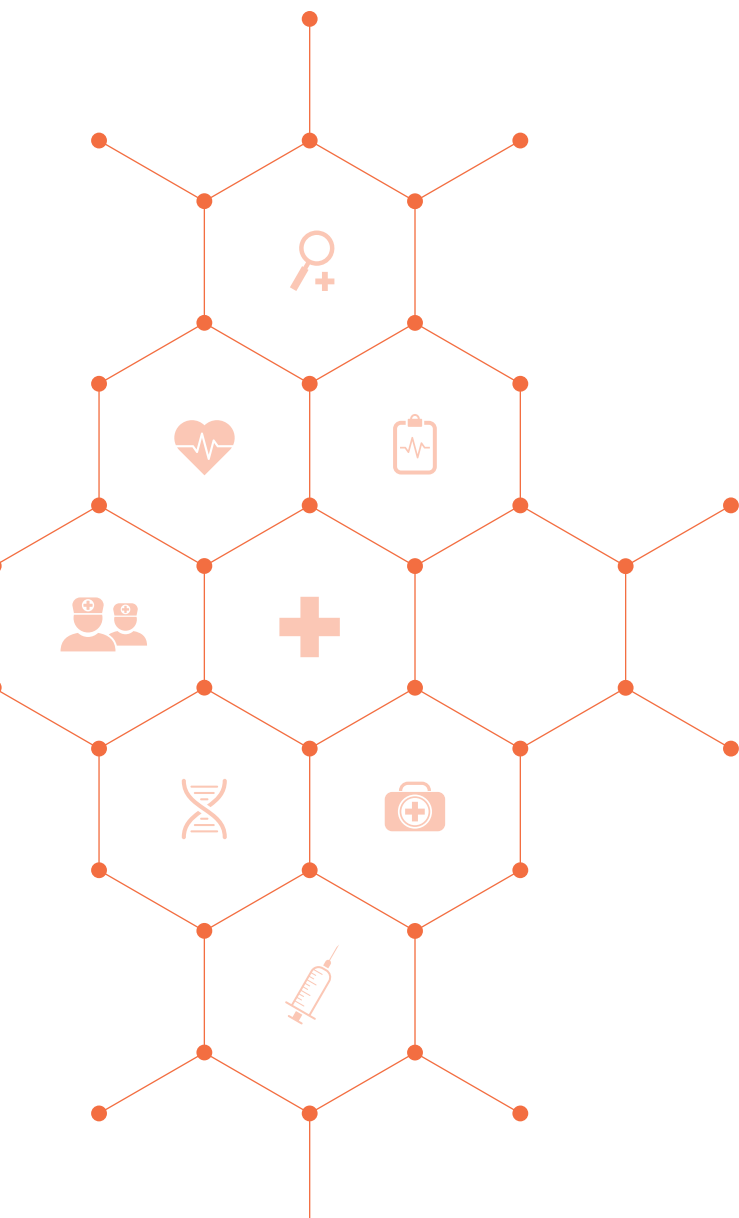
Pharmaceuticals

Leading pharmaceutical companies are using data science to develop more stable solutions for planning and conducting clinical trials.

Data science is helping pharmaceutical organizations predict and evaluate the success/failure ratio of their clinical trials, minimizing both risks and costs.

Many companies are also applying data science techniques to identify the right candidates for trials based on medical history,

body structure, and other vital characteristics.



REAL-LIFE EXAMPLES OF DATA SCIENCE

Here are some real-life examples of data science applications that most people use in their everyday lives, maybe without realizing it.



Internet Search Engine

All search engines, including DuckDuckGo, AOL, Bing, Yahoo, and of course, Google, use data science tools to provide personalized search results in seconds.

Given the fact that Google has been known to handle over 20 petabytes of data each day if data science didn't exist, the internet search giant would not have been what it is today.



Recommendations

We are all familiar with Facebook's friend suggestions, similar-product recommendations on Amazon, and individualized Netflix predictions based on past searches.

These companies, out of billions of alternatives, suggest or recommend only the products and services relevant to a particular user, enhancing the user experience. Data science makes this possible.

Image Recognition

When Facebook users upload their images with friends, Facebook instantly starts providing them with suggestions to tag their friends. By applying face recognition algorithms, data science powers this automated tagging feature on Facebook.

Be it Facebook, WhatsApp, Google, or any other application, data science is at the root of all image recognition techniques.

Speech Recognition

Voice-based services offered by Google Assistant, Apple Inc.'s Siri, Microsoft Cortana, and Amazon's Alexa are the best examples of speech recognition software.

These products, using data science, make life easier by enabling users to perform tasks by merely speaking out. The user's voice gets converted into text automatically.

DATA SCIENCE TERMINOLOGIES

If you are planning to start your career as a data scientist, mastering the key terminologies mentioned in this **basics of data science handbook** is essential to ensure success in your professional and educational path.

Familiarize yourself with the most important data science terminologies listed below.



Data Engineer

Data engineers develop the infrastructure that facilitates the collection, cleaning, and processing of data, which data scientists use to generate insights.



Machine Learning

Machine Learning (ML), an AI (Artificial Intelligence) subset, refers to techniques that data scientists apply to make computers (machines) learn from inputted data. ML techniques generate results without explicit programming rules.



Classification

Classification is a process to classify data into different classes. The purpose of Classification is to determine the class/category under which new datasets will fall.



Cross-Validation

Cross-Validation involves methods to validate the accuracy or stability of machine-learning models.



Clustering

Clustering refers to finding and segregating data points into groups that have similar traits.



Deep Learning

An advanced machine learning form that mimics the human brain, in-depth learning methodologies, based on ANN (Artificial Neural Network), can detect objects, translate languages, recognize speech, and make decisions from insights drawn from both unlabeled and unstructured data.



A/B Testing

A/B testing, a.k.a. split testing includes processes that compare versions of web pages, emails, or other digital assets, which helps measure performance differences.



Hypothesis Testing

Introduced by Karl Pearson, Ronald Fisher, and Jerzy Neyman, Hypothesis Testing refers to statistical methods that are used to make statistical decisions. It is often applied in clinical research.



EDA (Exploratory Data Analysis)

EDA techniques help summarize the main characteristics of datasets using visual methods. With Exploratory Data Analysis, data scientists can “see” what data tells them beyond hypothesis testing or formal modeling.



Data Visualization

Through visual elements, such as maps, charts, and graphs, data visualization offers a graphical portrayal of data that helps data scientists visualize and understand data trends, patterns, and outliers.



Data Modeling

A process to produce descriptive diagrams of linkages between different pieces of information stored in databases, data modeling is among the key skills that data scientists must be proficient in to do research design and data store architecting.



Data Warehouse

A core component of data-driven businesses, data warehouses are relational databases that contain historical transaction data for analysis and query. Data warehouses incorporate myriad frameworks and tools that work holistically to make the data available for extracting insights.



MUST-HAVE DATA SCIENCE SKILLS TO GET HIRED

Because of the present talent gap, data science is, at the moment, the most in-demand and promising career option for professionals with the right skill-sets.

Newcomers in data science, besides having in-depth knowledge of mathematics, statistics, and programming languages such as Python, R, and SQL, should also have industry-specific skills, including:



Machine Learning

As a data scientist working for a large organization that generates a massive amount of data, you should be well-versed in various machine-learning techniques, including K-Nearest Neighbors, Ensemble Methods, and Random Forest.

You can implement most of these techniques using Python or R, but more importantly, you must be able to understand and determine when to apply these different machine-learning techniques.



Data Wrangling

In the early stages of your career as a data scientist, you will not only be responsible for data analysis, but your job role may also involve cleaning up dirty, imperfect, and messy datasets.

You would have to be good at dealing with data imperfections, including timestamps versus unix time, missing values, and inconsistent string and date formatting, such as 'new york' v/s 'New York', or '01/01/2020' v/s '2020-01-01'.



Data Visualization and Communication Skills

Most organizations hire data scientists to boost their decision-making capabilities.

To help businesses make data-driven decisions, you need to be a specialist in data visualization with a thorough knowledge of data visualization dashboards and tools, including ggplot, d3.js, Matplotlib and Tableau. You should also have excellent communication skills to share your findings with all stakeholders.



Critical Thinking

In data science, critical thinking skills enable you to approach problems from diverse perspectives. It also empowers you to analyze results, questions, and hypotheses effectively, which are crucial to solving problems in the real world.

Developing critical thinking skills is vital because you, as a data scientist, not only have to find insights, but you also need to articulate relevant questions to understand how results translate into actions required to take business-critical steps.



Problem-Solving

The ability to solve problems is one of the key roles of a data scientist. You cannot become a successful data science professional without the will and skill to solve critical problems.

Top-notch data scientists are outstanding problem-solvers who can quickly identify problems and then find the most appropriate methods to get the right solutions to those problems with the resources available.

SOME INTERESTING STATS ON DATA SCIENCE

Data Scientist: The Sexiest Job of the 21st Century—**Harvard Business Review**

The World Economic Forum **forecasts** that data scientists will emerge as the number one role in the world by **2022**.

According to **LinkedIn**, there has been a **650 percent job growth** in this field since 2012.

A report by the **U.S. Bureau of Labor Statistics** states that there will be **11.5 million new job openings** by 2026.

Job-search platform **Glassdoor** shows that the average salary of a data scientist in the United States is **\$1,13,309** per annum.

All of this sounds divine, but where do you learn the key data science skills needed to land a high-paying job?



START YOUR JOURNEY TO BECOMING A DATA SCIENCE EXPERT

Now that you have a comprehensive overview of the field of Data Science, the career opportunities that await you, and the skills you need to get there, the next and most effective step towards achieving your goal is to get certified and learn all you need to. Simplilearn is a pioneer in online training and one of the world's leading certification providers in the most in-demand technologies today. We provide various training and certifications, for all levels of professionals (beginners to senior level) to equip you with the knowledge required to forge a career path in data science.



Basic Courses

- ✓ **Data Science Certification Training - R Programming**
- ✓ **Data Science with Python**



Master's Program

- ✓ **Data Scientist Master's Program**



Post Graduate Program

- ✓ **Post Graduate Program in Data Science**

Start scripting your career success story today.



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